

KHULEKANI MTSHALI

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience across software development, IT infrastructure, systems support, and cloud environments, and a growing interest in artificial intelligence and its real-world applications. Proven ability to diagnose complex technical problems, maintain secure and high-availability systems, and communicate clearly with both technical and non-technical stakeholders. Comfortable working across the full technology stack, from hardware and OS-level troubleshooting to application development, database management, and cloud deployment. Passionate about leveraging AI and emerging technologies to drive innovation and deliver business value. Seeking roles in software engineering, IT operations, cloud computing, data, AI, cybersecurity, or technical support where I can add value through structured problem-solving and a strong technical foundation.

EDUCATION

BSc Computer Science, University of the Witwatersrand 2022 – 2025

Data Structures & Algorithms · OOP · Software Engineering · Operating Systems · Computer Networks · Database Systems · System Analysis & Design · Cybersecurity Fundamentals · Cloud Computing · Web & Mobile Development · Human-Computer Interaction · Organisational Data Management

National Senior Certificate, Luthuli High School 2021

TECHNICAL SKILLS

Programming & Development: Python, Java, C++, JavaScript: OOP, data structures, algorithms, complexity analysis, system design

Operating Systems: Windows & Linux: configuration, administration, troubleshooting, performance tuning

Networking & Infrastructure: TCP/IP, DNS, DHCP, IP configuration, connectivity diagnostics, managed lab infrastructure

Databases & Data: MySQL, Firebase: schema design, querying, data integrity, basic data modelling

Cloud Platforms: AWS EC2, S3, IAM: deployment, access management, cloud fundamentals (AWS-certified); Microsoft Azure AI services

AI & Machine Learning: Supervised & unsupervised learning, generative AI, prompt engineering, recommender systems, reinforcement learning; tools: ChatGPT, Claude, Copilot, Gemini

Cybersecurity: Access control, endpoint security, secure configuration baselines, permission management

IT Support & Systems: Tier 1 & Tier 2 support, workstation imaging, patch management, root cause analysis, incident documentation

Tools & Collaboration: Git, GitHub, Linux CLI, Windows CLI, CRM & ticketing systems

WORK EXPERIENCE

End-to-End Technology Practitioner | Capaciti

April 2026 – Present

- Enrolled in a structured end-to-end technology programme, gaining practical exposure across the full technology delivery lifecycle.
- Applying skills in software development, cloud computing, and IT systems within a professional capacity-building environment.
- Collaborating with peers and industry mentors on technology projects designed to develop job-ready competencies.

Laboratory Technical Assistant | University of the Witwatersrand

2023 – 2026

- Delivered Tier 1 and Tier 2 technical support across multi-user lab environments, resolving OS, application, and network incidents with minimal downtime.
- Deployed, configured, and maintained Windows and Linux workstations including IDEs, compilers, database engines, and virtualisation platforms.
- Conducted system-level diagnostics and root cause analysis on hardware faults, driver conflicts, OS corruption, and performance degradation.
- Troubleshoot TCP/IP connectivity, DNS failures, authentication errors, and network misconfigurations within managed lab infrastructure.
- Executed workstation imaging, patch management, and controlled software deployment to maintain configuration consistency and security compliance.
- Applied access control, permission management, and endpoint hygiene standards across shared multi-user systems.
- Documented technical incidents and implemented preventative controls to reduce recurring failures and improve system reliability.

Call Centre Agent | DIAL-N-DINE, LLC

2023 – 2025

- Managed high-volume inbound operations using structured ticketing and CRM systems under defined SLA constraints.
- Diagnosed and resolved transactional and system-level issues, escalating technical anomalies to appropriate support tiers.
- Maintained confidentiality and integrity of sensitive customer data in compliance with data handling standards.
- Identified recurring system inefficiencies and contributed to process improvement recommendations.

First-Year Experience Mentor | University of the Witwatersrand

2024 – 2025

- Provided structured onboarding guidance, assisting students in navigating institutional digital systems and academic workflows.
- Supported adoption of technical coursework through structured mentoring, resource alignment, and one-on-one coaching.

CERTIFICATIONS & TRAINING

Trustworthy AI: Managing Bias, Ethics, and Accountability, The Johns Hopkins University / Coursera | June 2026

Unsupervised Learning, Recommenders, Reinforcement Learning, DeepLearning.AI / Coursera | June 2026

Supervised Machine Learning: Regression and Classification, DeepLearning.AI / Coursera | June 2026

Artificial Intelligence on Microsoft Azure, Microsoft / Coursera | June 2026

Google AI Essentials Specialization, Google / Coursera | June 2026

Python for Data Science, AI & Development, IBM / Coursera | May 2026

AI Essentials, Intel / Coursera | May 2026

Introduction to Generative AI, Google / Coursera | May 2026

Generative AI: Prompt Engineering Basics, IBM / Coursera | May 2026

Introduction to Artificial Intelligence (AI), IBM / Coursera | May 2026

AI For Everyone, DeepLearning.AI | April 2026

Full Stack Development (32 Credits), FNB App Academy | July 2025

AWS Cloud Bootcamp, CloudSec Network | September 2025

KEY STRENGTHS

Adaptable across technical disciplines · Strong written and verbal communication · Structured problem-solving & root cause analysis · Works well under pressure and SLA constraints · Experienced in both independent and team-based environments · Quick learner with a track record of upskilling

REFERENCES

Available on request.